

FLOWTRACK

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Executive Summary

FlowTrack is a multi-tenant SaaS platform for time tracking, activity monitoring, invoicing, and workforce operations. The product is feature-rich and functional. The remaining work is production hosting, desktop code signing, and security review.

QUESTION 1

What is the complete technology stack and architecture?

ARCHITECTURE — 3-TIER MULTI-TENANT SAAS

COMPONENT	ROLE	TECHNOLOGY
Desktop Tracker	Real-time timer, app/browser tracking, screenshots	Electron 31
Web Application	Dashboard, reports, settings, marketing site	React 19 + TypeScript
Backend API	Business logic, auth, billing, integrations	PHP 8.1+ / CodeIgniter 4
Database	Tenant-scoped persistent storage	MySQL 8+

COMPLETE TECHNOLOGY STACK

LAYER	TECHNOLOGIES
Backend	PHP 8.1+, CodeIgniter 4, Composer
Database	MySQL 8+ (MariaDB compatible)
API authentication	JWT (access + refresh), OAuth, 2FA (Google Authenticator)
Frontend	React 19, TypeScript, Vite 6, Tailwind CSS 4
Desktop	Electron 31, active-win (foreground app), electron-updater
Payments	Stripe (subscriptions, webhooks, platform ledger)
Documents	DomPDF (PDF), PhpSpreadsheet (Excel exports)
Integrations	Slack, Microsoft Teams, Jira, GitHub, Google/Outlook Calendar

SECURITY & TENANCY MODEL

- Each customer is an **organization**; all data scoped by `organization_id`.
- **RBAC** with granular permissions: Owner, Admin, Manager, Team Lead, Member.
- **Plan-based feature gating** enforced at API level (screenshots, activity, invoicing, etc.).
- JWT auth filter on all protected routes; permission filter per endpoint.

QUESTION 2

What is the current status of the product?

Below is a precise breakdown of what is fully working today and what still needs development before public launch.

✓ FULLY WORKING — CORE PRODUCT

Time Tracking

Start, pause, resume, stop timer (web + desktop) · Manual entries · Active session sync · Midnight split & orphaned timer cleanup

Desktop Tracker (FlowTrack Tracker)

Windows installer + auto-update · Foreground app detection · Browser tab/site breakdown (TikTok, YouTube, Chrome) · Screenshot capture & cloud sync · Idle detection · Offline queue · Daily summary UI

Activity & Productivity

Activity log sync from desktop · Productivity rules (app, URL, keyword) · Productive/unproductive/neutral categorization · Bulk re-categorization · Team activity feed & member monitoring

Organization & Access Control

Multi-tenant organizations · Member invitations · Role assignment · Granular permissions · Plan feature flags · Per-member monitoring settings

Projects & Tasks

Project CRUD, archive, member assignment · Tasks linked to projects · Time entries associated with projects/tasks

Timesheets & Leave

Weekly timesheet generation · Submit, approve, reject workflow · Leave types (annual, sick, casual, unpaid)

Screenshots

Plan-based capture intervals · Thumbnail gallery, pagination, full preview · Blur/sensitivity policies · Signed/JWT-protected image URLs

Reports & Analytics

Dashboard · Hourly timeline (productive/unproductive/neutral/idle) · Top apps & browser tabs · Time summary · Team analytics · Advanced monitoring reports

Billing & Subscriptions

Tiered plans with feature limits · Stripe checkout, subscriptions, webhooks · Trial management, coupons · Organization billing page · Admin revenue dashboard

Invoicing, Payroll & Client Trust

Invoice generation from tracked time · Client portal (review, approve, pay) · Proof-of-Work pack with integrity score · Verifiable work certificate · Payroll runs & payslip PDF export

Integrations

Slack (commands, notifications, hub) · Microsoft Teams (commands) · Jira (issue linking, hub) · GitHub (commit context, hub) · Google/Outlook Calendar sync

AI & Insights

AI activity categorization · Daily standup generation · Ask FlowTrack / insights panels · Unusual activity detection · Wellbeing & burnout signals · Predictive forecasting hooks

Platform Admin (Super-Admin)

Organization & user management · Plan & feature management · Revenue, payments, coupons · Marketing campaigns · System health, audit log · User impersonation

Marketing & Onboarding

Public landing page with pricing · Registration & email verification · Onboarding checklist · Privacy policy & terms pages

Development Quality

GitHub Actions CI (PHPUnit + frontend lint/build) · Health endpoint · Release checklist & incident runbooks

STILL NEEDS WORK BEFORE PUBLIC LAUNCH

AREA	CURRENT STATE	REQUIRED BEFORE LAUNCH
Production hosting	...	Dedicated server, SSL, custom domain
Desktop code signing	Windows unsigned; macOS CI documented	Apple notarization; Windows signing
Scheduled jobs	CLI commands implemented	Production cron (timer sweep, campaigns)
Automated testing	Basic PHPUnit only	API integration & E2E tests
QA & regression	Core flows verified in dev	Full pass across roles, plans, clients
Advanced AI UX	Backend services exist	Polish flagship AI timesheet experience

Launch Readiness Checklist

READY FOR BETA

- ✓ Core time tracking (web + desktop)
- ✓ Activity & screenshot monitoring
- ✓ Team management & permissions
- ✓ Reports & analytics
- ✓ Billing (Stripe) & plan limits
- ✓ Windows desktop installer

In Short

Q: What is the complete technology stack and architecture?

FlowTrack is a **3-tier multi-tenant SaaS**: Electron desktop tracker + React web app → CodeIgniter 4 REST API → MySQL. Auth uses JWT, OAuth, and 2FA. Payments via Stripe. Integrations include Slack, Teams, Jira, GitHub, and Calendar. All data is organization-scoped with role-based permissions and plan-based feature gating.

Q: What is fully working and what still needs development before launch?

Working: Time tracking, desktop app (timer, app/browser tracking, screenshots), activity logs, productivity rules, projects/tasks, timesheets, leave, reports, invoicing, payroll, client portal, proof-of-work, Stripe billing, integrations, AI insights, and super-admin panel.

Before launch: Production hosting, desktop code signing, production cron jobs, full QA, security review, and API documentation polish. Product is ~85–90% complete; beta is feasible now; public launch estimated at 4–6 weeks.

METRIC	STATUS
Product completeness	~85–90%
Launch readiness	~70–75%
Beta (limited users)	Feasible now on staging
Public commercial launch	~4–6 weeks with focused infra & QA

Note: FlowTrack is not a prototype — it is a feature-rich, working product. The remaining work is production deployment, quality assurance, desktop trust (code signing), and polish.